Date: June 6th, 2026

Week 1: Research Fundamentals (Academic Papers & Preliminary Proposal)

CariSurg Healthcare AI Program

TASK 1 — Literature Review (ONLY the 3 provided papers)

Paper 1 — Use of Artificial Intelligence in Triage in Hospital Emergency Departments (Scoping Review)

This scoping review examines 29 studies on AI applications in emergency department triage. The central problem is inconsistent and resource-constrained triage decision-making in high-volume ED settings. Methods across included studies involve machine learning approaches such as logistic regression, random forest, gradient boosting, and deep neural networks applied to clinical and administrative ED data. Findings show AI improves triage accuracy, reduces under- and over-triage, and supports prediction of admission, ICU need, and mortality. Limitations include retrospective study designs, heavy reliance on single-centre datasets, and a lack of prospective real-world validation.

Paper 2 — Araouchi & Adda (2024), ICTH Conference

This study addresses the challenge of predicting KTAS triage levels in overcrowded emergency departments. The problem is variability and workload burden in manual triage classification. Methods include supervised machine learning models (SVM, RF, ANN, GBM, XGBoost, LR) and a stacking ensemble trained on 1,267 ED patient records. The stacking model achieved the best performance (~80% accuracy, AUC ~0.93). Results show ensemble learning improves predictive accuracy for triage classification. Limitations include small sample size, single-country data, and reduced generalisability across different healthcare systems.

Paper 3 — Raita et al. (ED Machine Learning Outcome Prediction Studies)

This study evaluates machine learning models for predicting emergency department outcomes such as ICU admission and mortality. The clinical problem is early identification of high-risk patients during triage. Methods include logistic regression, random forests, gradient boosted decision trees, and deep neural networks using ED clinical and administrative datasets. Results show ML models outperform traditional scoring systems in predictive performance. Limitations include limited external validation, retrospective design, and difficulty integrating models into real-time emergency department workflows.

TASK 2 — Gap Analysis (based ONLY on these 3 papers)

Gap 1 — Lack of real-time, prospective validation in ED workflows

All three papers evaluate models using retrospective datasets. None test AI triage systems in live emergency department conditions where clinicians make decisions under time pressure, missing data, and operational constraints. This limits understanding of real-world safety, usability, and clinical impact.

Gap 2 — Limited external generalisability across hospitals and populations

Both Paper 1 and Paper 2 rely heavily on single-centre or limited population datasets. Paper 3 also highlights challenges in external validation. There is insufficient evidence that AI triage models maintain performance across different hospitals, patient demographics, and healthcare systems.

TASK 3 — Preliminary Problem Statement (≤ 90 words)

Emergency Departments rely on manual triage systems that are vulnerable to variability under high workload and time pressure. Evidence from recent studies shows that AI models can improve triage accuracy and outcome prediction; however, current evaluations are largely retrospective and lack validation in real clinical environments. This creates uncertainty regarding safety, reliability, and clinician trust in operational use. This 12-week pilot will evaluate a machine learning-based triage support tool in a shadow deployment within Mercer ED workflows, assessing concordance with clinician decisions, feasibility of integration, and early performance under real-world constraints.